Politecnico di Milano
Dipartimento di Elettronica, Informazione e Bioingegneria
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Parallel Computing– II part– Thursday, June 13^{th}, 2024

Polimi ID ________ ____________________________

Surname ________ ______________________ **Name** _________________________

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _________________________

Exercise 2 (4 points) _________________________

Exercise 3 (4 points) _________________________

Exercise 3 (4 points) _________________________

Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how the parallel gather pattern works. (2)

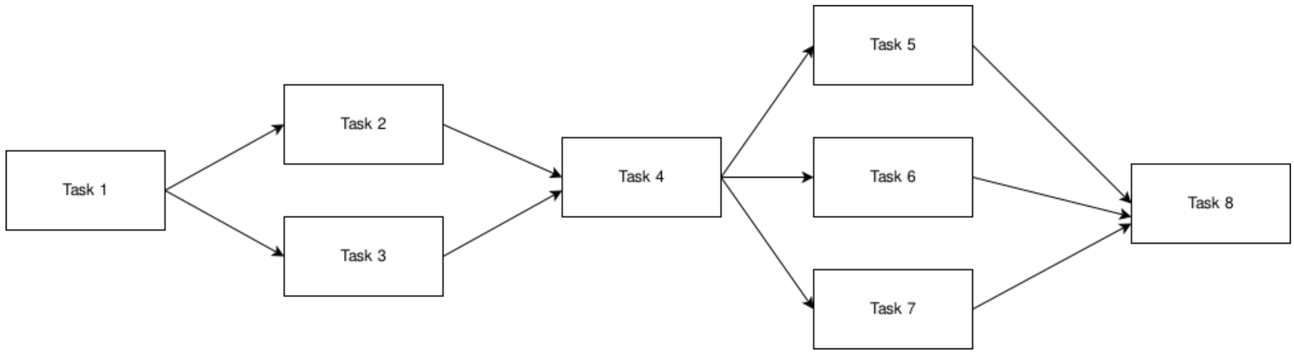
B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=0; i<100; i++)
        for (j=i; j<100; j++)
           a[i][j] = f(a[i][j]);
```

Exercise n. 2

Answer the following questions about OpenMP (without an explanation, the answer will be considered invalid).

A. Consider the following task graph:



Where: W(Task 1) = 50, W(Task 2) = 100, W(Task 3) = 100, W(Task 4) = 20, W(Task 5) = 10, W(Task 6) = 150, W(Task 7) = 175, W(Task 8) = 50

Is it a good idea to implement the program using three threads? What is the improvement with respect to a sequential version? (1)

Write a corresponding OpenMP implementation. (2)

B. What is the difference between OpenMP sections and tasks? (1)

Exercise n. 3

Answer the following questions about CUDA (without an explanation, the answer will be considered invalid).

A. Briefly describe the three different ways in which it is possible to access memory across different CUDA devices. (2)

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B. Describe the types of memory available on NVIDIA GPUs and who can access each of them. (1)

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C. What are thread hierarchy variables in CUDA and why are they useful? (1) **This question can be skipped in case of a passed challenge.**

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Exercise n. 4

Answer the following questions about parallel patterns in CUDA (without an explanation, the answer will be considered invalid).

A. For the **work-inefficient** scan kernel based on reduction trees, assume that we have 2048 elements in the input vector, explain how to estimate the number of add operations performed. (2)

B. For the **work-efficient** scan kernel based on reduction trees and inverse reduction trees, assume that we have 2048 elements in the input vector, explain how to estimate the number of add operations performed in both the forward and the reverse phase. (2)